



Silicate Carbon

Carbon dioxide removal prepurchase application 2025

General Application

(The General Application applies to everyone; all applicants should complete this)

Public section

The content in this section (answers to questions 1(a) - (d)) will be made public on the [Frontier GitHub repository](#) after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Company or organization name

Silicate Carbon Limited

Company or organization location (we welcome applicants from anywhere in the world)

Dublin, Ireland

Name(s) of primary point(s) of contact for this application

Maurice Bryson

Brief company or organization description <20 words

Silicate Carbon accelerates silicate and carbonate mineral weathering to durably sequester atmospheric carbon dioxide and amend soil pH in farmland.

1. Public summary of proposed project¹ to Frontier

- a. **Description of the CDR approach:** Describe how the proposed technology removes CO₂ from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. If your project addresses any of the priority innovation areas identified in the RFP, tell us how. Please include figures and system schematics and be specific, but concise. 1000-1500 words

¹ We use "project" throughout this template, but the term is not intended to denote a single facility. The "project" being proposed to Frontier could include multiple facilities/locations or potentially all the CDR activities of your company.

Introduction

Silicate Carbon has pioneered field-scale terrestrial enhanced weathering (EW) science since 2021. Working originally with silicate- and carbonate-bearing returned concrete, its foundational research was published in 2024¹. Over time, the company expanded its focus to carbonate minerals in general, utilizing calcium carbonate (CaCO₃) as its primary material. The company has relentlessly pushed EW science forward by developing a very large database of shallow soil-water chemistry across multiple continents; it also leads the world's first long-term continuous greenhouse gas (GHG) flux trial in EW, and is advancing novel, proprietary, high-precision isotopic methods to unlock carbon dioxide removal (CDR) measurement at scale.

Silicate Carbon is headquartered in Dublin, Ireland, with a U.S. subsidiary in Chicago, Illinois. The company's operations center around Wexford, Ireland, and Kankakee County, Illinois, U.S., where it has applied ~10,000 tons (t) of pH amendment materials to ~1,000 hectares (ha) of arable land since 2021. This work, coupled with the project outlined here, provides the data, tools and operational insights necessary to unlock gigaton (Gt) CDR through EW.

Carbon dioxide removal and (>1,000-year) storage concept

The chemical weathering of calcium (Ca)- and magnesium (Mg)-bearing silicate rocks and minerals consumes atmospheric carbon dioxide (CO₂) and regulates Earth's climate over geologic timescales². Over shorter human timescales, the chemical weathering of carbonate minerals, such as CaCO₃, serves as a more rapid sink for atmospheric CO₂³. Carbonate weathering accounts for 94% of the 947-1,057 megatons (Mt) CO₂ sequestered by background weathering each year^{4,5}, because carbonate minerals dissolve ~10⁴-10⁶ times faster than silicate minerals⁶. Equations (1) and (2) provide generalized examples of silicate and carbonate weathering, respectively:



Carbonic acid (H₂CO₃) dissolves rock-forming minerals, such as CaCO₃ to convert CO₂ to bicarbonate (HCO₃⁻). Surface and ground waters transport HCO₃⁻ to the ocean (Figure 1), where it has a residence time of ~80,000 years⁷. In this way, rock weathering reduces the concentration of planet warming CO₂ in Earth's atmosphere, and it forms the basis of Silicate Carbon's Frontier prepurchase application.



Figure 1 | Process flow diagram of Silicate Carbon's EW activities. Following pulverization and soil amendment, chemical weathering of minerals releases alkalinity in the form of HCO₃⁻, which is transported via surface and ground waters to the ocean where it resides for ~80,000 years before precipitating as CaCO₃⁷.

Differentiation

Silicate Carbon is unique among commercial EW efforts because it harnesses the immense CDR potential of carbonate minerals. These naturally occurring minerals have several advantages vis-à-vis silicate minerals, including fast weathering rates, global abundance, proximity to farmland (Figure 2), low cost, and an established use in agricultural soil enhancement. The CDR process described here, once scaled above business-as-usual (BAU) practices, could theoretically remove an additional 0.94 Gt-1.9 Gt CO₂ y⁻¹ (unpublished) beyond current carbonate-weathering CDR, while improving the productivity of farmland and increasing the resilience of agriculture to financial and climate stresses. While not included in this scenario, a study modelling global liming impacts estimated that CDR potential could increase from 0.61 Gt CO₂ y⁻¹ to 3.09 Gt CO₂ y⁻¹¹⁸.

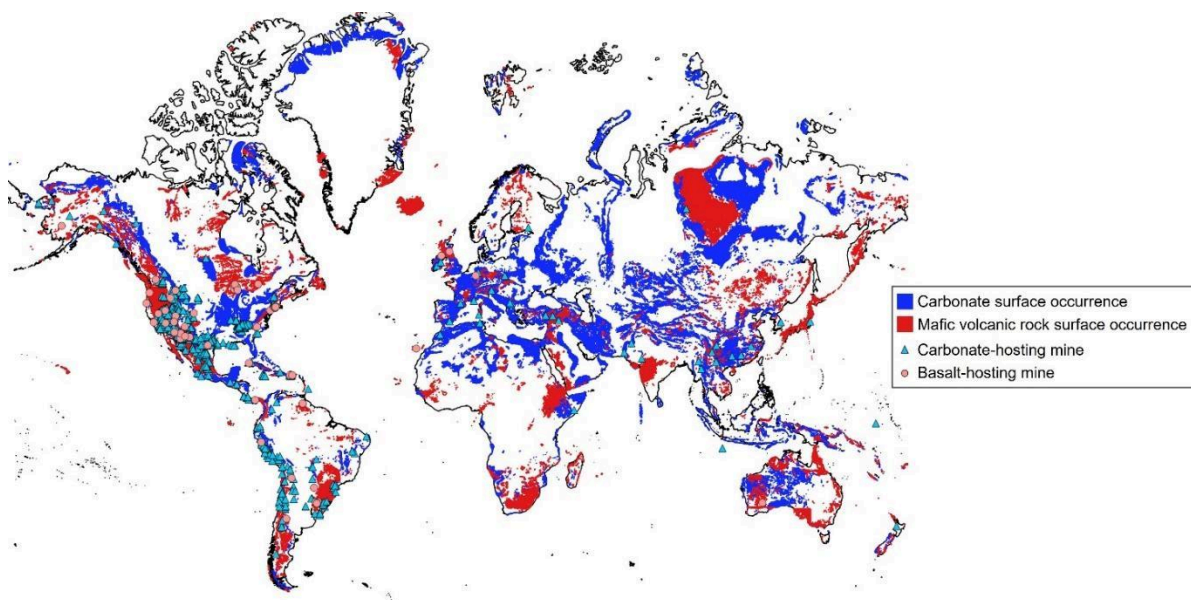


Figure 2 | Global lithological map showing surface coverage of carbonate sedimentary rocks and basic volcanic rocks, as mapped and defined by the Global Lithological Map Database⁹.

More specifically, Silicate Carbon prioritizes carbonate weathering, rather than silicate weathering, in the mid-latitude agricultural regions where it currently operates, because carbonate minerals dissolve 10⁴-10⁶ times faster than silicates⁶, do not contain hazardous metals (e.g., nickel and chromium), dissolve efficiently in temperate conditions¹⁰, and do not form secondary clay minerals¹¹. Silicate Carbon has demonstrated that CDR can be confidently measured in soil waters following the application of carbonate minerals at relatively low amendment rates¹.

Best-in-class approach

Since undertaking the world's first EW field trial with carbonate minerals in 2021¹, Silicate Carbon has expanded its research and operations to multiple geographies. Among several milestones of note, the company was selected to join the Breakthrough Energy Fellows program in 2023, became an XPRIZE Carbon Removal finalist in 2024, and was awarded funding under the U.S. Department of Energy's Carbon Negative Shot program in 2024, alongside academic collaborators Northwestern

University and University College Dublin. The team's success weathering carbonate minerals to efficiently sequester atmospheric CO₂ is typified by the team's approach to measuring CDR. Silicate Carbon prioritizes direct sampling of HCO₃⁻ in situ, supported by CO₂, nitrous oxide (N₂O) and methane (CH₄) soil-air gas flux monitoring, and novel, proprietary isotopic methods to constrain Ca export in the solid phase. The merits of Silicate Carbon's approach to EW that make it best in class include:

- **Efficient CDR**

- Optimal field conditions are paramount to efficient CDR through EW. Silicate Carbon's proprietary field selection tool coupled with the rapid dissolution kinetics of its carbonate feedstock material, CaCO₃, ensure robust and directly measurable increases in dissolved HCO₃⁻. Silicate Carbon selects fields and optimizes material application according to several factors determined over four years of extensive field-scale research that collectively yield demonstrable increases in HCO₃⁻ alkalinity production and, importantly, export to the ocean. Factors such as process emissions and downstream losses are also considered during the field selection phase, ensuring low-cost, efficient net CDR.

- **Measurement**

- Scientific consensus surrounding CDR measurement for EW has not been achieved. At a minimum, it is necessary to measure HCO₃⁻ alkalinity, as this is the carbon (C) sequestered from the atmosphere. Measurement strategies employing the combined analysis of rocks, soils, and waters are needed to confidently constrain CDR. Silicate Carbon uses redundant and independent measurement techniques to offer best-in-class measurement of CDR through EW, specifically across soil water, soil-air GHG flux, and soil measurements. Silicate Carbon has broad operational expertise sampling all phases of the weathering process at scale—a video detailing the team's measurement process can be seen [here](#).

- **Scale**

- The global availability of limestone and related carbonate sedimentary deposits is considered vast and widely distributed, comprising ~8% of all surface rocks (Figure 2)⁹. Annual limestone production is estimated to be more than 6.6 Gt¹², accounting for 9% of the world production of mineral raw materials (~44 Gt y⁻¹). Importantly, limestone deposits are proximal to global crop fields¹³, meaning transport-related emissions for EW can be minimized, ensuring efficient net CDR.
- Silicate Carbon has robust partnerships across agriculture, mineral extraction/processing, and analytics, developed over several years of large-scale EW deployments in the U.S. and Europe. These partnerships will enable rapid expansion of carbonate EW. Silicate Carbon has a fully equipped laboratory in-house in Ireland and the U.S. to ensure rapid and robust CDR measurement data acquisition, permitting efficient scaling.

Priority innovation areas

Diversifying material selection, deployment areas and measurement techniques will increase the probability of Gt CDR y⁻¹ with EW. To date, the industry has focused on silicate-bearing materials, such as basalt and olivine, primarily due to Goldich (1938)¹¹, as well as the fact that carbonate weathering is not a CO₂ sink over geological timescales^{2,11}. However, over timescales relevant to the immediate climate crisis (i.e., annual-decadal), carbonate weathering is a more efficient CO₂ sink due to the vast difference in dissolution kinetics between silicates and carbonates³⁻⁵. Importantly, HCO₃⁻ produced from carbonate weathering can be stable in oceans for ~80,000 years⁷, meaning that carbonate EW

offers both rapid and durable CDR¹⁴.

Another important area of diversification in EW is CDR measurement. Solid phase assessments of cation loss from feedstocks, such as 'TiCAT', have been suggested as a method of estimating weathering¹⁵⁻¹⁷ and have become common in the industry. However, this method fails to track CDR directly, quantifying instead cation loss from applied feedstocks. Failing reconciliation of solid-phase measurements with soil water alkalinity, such approaches could overestimate CDR due to misinterpreting strong acid weathering as H_2CO_3 weathering¹⁸⁻²⁰. Silicate Carbon's expertise in directly quantifying CDR through soil water chemistry in large field-scale deployments could mitigate these concerns and further enhance efforts to constrain CDR in EW.

Conclusion

Silicate Carbon has transformed EW from a theoretical CDR pathway into a measurable, scalable solution by directly quantifying the journey of C from atmospheric CO_2 to HCO_3^- . The team's high-frequency soil water sampling approach directly measuring HCO_3^- , combined with comprehensive laboratory analyses, and redundant, independent measurements across the solid and gaseous phases, position Silicate Carbon at the forefront of EW research and deployment.

The company's field-scale research over four years confirms limestone as a material with Gt CDR potential, warranting its inclusion in a diverse portfolio of CDR pathways. Moreover, the material's established presence in agricultural supply chains and proven soil benefits provide immediate pathways to scale.

Critically, Silicate Carbon has acquired a large, robust and comprehensive database of shallow, soil water chemistry data across many fields in multiple geographies that supports limestone's CDR potential. These data, combined with Silicate Carbon's proprietary field selection and CDR measurement tools, among other proprietary assets, combine to enable effective, efficient and scientifically robust CDR through EW. With Frontier's support, Silicate Carbon will accelerate the deployment of EW as a vital tool for addressing the climate crisis using a solution that can be implemented at scale, today.

- b. **Project objectives:** What are you trying to build? Discuss location(s) and scale. What is the current cost breakdown, and what needs to happen for your CDR solution to approach Frontier's cost and scale criteria?² What is your approach to quantifying the carbon removed? Please include figures and system schematics and be specific, but concise. 1000-1500 words

Company objectives

Silicate Carbon is expanding carbonate EW across farmland, globally, to reach Gt CDR before mid-century. From field work to date, the company projects a potential gross CDR of 0.94 Gt-1.9 Gt $\text{CO}_2 \text{ y}^{-1}$ from enhanced carbonate mineral weathering on farmland (unpublished).

² We're looking for approaches that can reach climate-relevant scale (about 0.5 Gt CDR/year at \$100/ton). We will consider approaches that don't quite meet this bar if they perform well against our other criteria, can enable the removal of hundreds of millions of tons, are otherwise compelling enough to be part of the global portfolio of climate solutions.

Silicate Carbon is initially targeting regions where limestone is underutilized to amend soil pH, but where infrastructure to pulverize, deliver and apply material is available. Those regions include parts of Ireland and the U.S. Midwest. The company will expand to other parts of Europe, Africa, North America, South America and Asia as the CDR process becomes more established.

By leveraging existing limestone quarries and processing facilities, while targeting agricultural soils that require soil pH amendment, Gt CDR y^{-1} can be achieved rapidly. The economic and geochemical understanding, as well as the widespread abundance, low cost, and known ecological impacts of limestone, make it a CDR process with merit both now and at the Mt and Gt scale.

Project overview

The Frontier prepurchase project will validate a previously overlooked mineral group for EW, carbonates. It will measure CDR at scale using shallow soil water chemistry as the primary indicator of CDR, with supporting data from the solid and gaseous phases, as well as downstream measurements to constrain potential HCO_3^- losses and potential impacts to BAU CO_2 degassing. It will incorporate novel insights from hydrologically isolated field-scale assessments, further supporting a best-in-class approach to EW MRV.

To the proposer's knowledge, no largescale EW project has, at scale, (i) selected candidate fields based on soil characteristics' impact on HCO_3^- production and export, (ii) sampled soil water at depths >30 cm, and (iii) continuously measured gaseous CO_2 concentrations and C isotopes, along with corresponding N_2O concentrations. As such, the proposed project will offer novel insights on essential determinants of CDR efficacy that will be relevant to all proposed EW materials across all geographies. Importantly, this work will expand the portfolio of rocks and measurement methods available to EW, further enhancing the probability of Gt CDR y^{-1} being achieved rapidly. The results from this work will be made public through peer-reviewed academic papers.

Silicate Carbon's expertise in field selection, soil amendment, and CDR measurement will be applied at a 650-ha scale project in SE Ireland, developing robust EW CDR measurements constrained by multiple, independent measurement methods. Building on the team's four years of experience effectively applying materials and measuring CDR in EW, this work will unlock vital evidence establishing the efficacy of (i) carbonate minerals for rapid CDR, and (ii) the utility of soil water chemistry for constraining CDR with EW at scale. The project will serve as an incremental yet vital step towards even larger deployments, setting the foundations for the expansion of limestone soil enhancement to improve soil productivity through soil pH amendment, and to rapidly sequester CO_2 as HCO_3^- .

Project location

The project will be undertaken in southeast Ireland. This area represents ideal conditions for an EW project due to its high levels of rainfall, as well as a base lithology and soils that provide optimal conditions for HCO_3^- production and export. The rivers draining this area are well understood from previous Silicate Carbon research, indicating that calcite precipitation (i.e., CO_2 loss) during downstream transport of EW products is unlikely. However, these downstream impacts will be measured, and form part of the MRV process of this project. Figure 3 illustrates the area to be employed for the proposed Frontier prepurchase project.

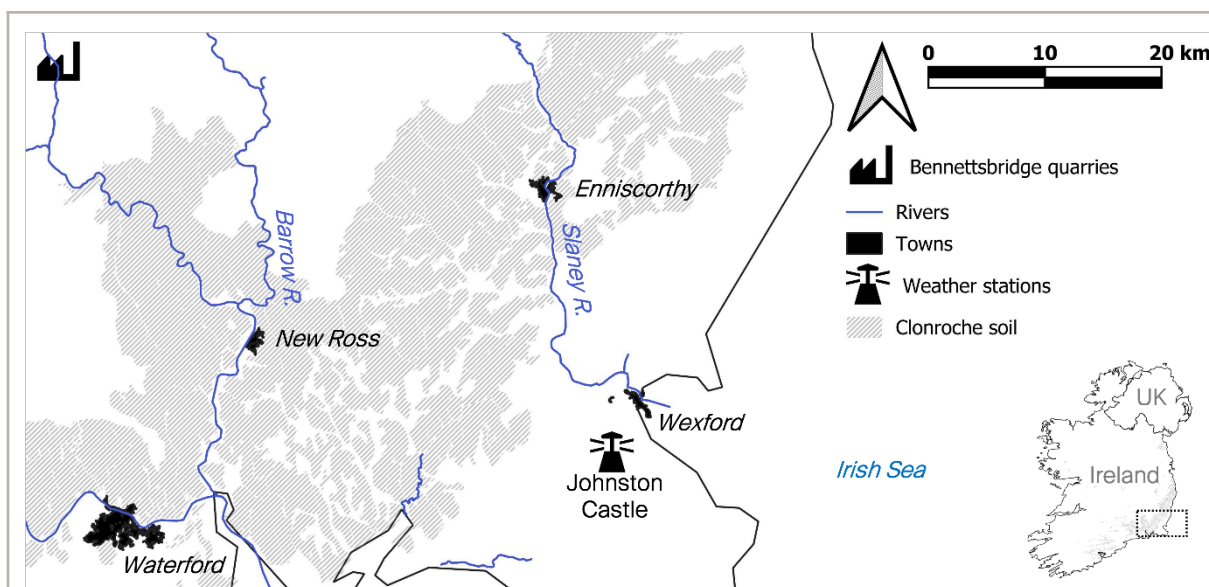


Figure 3 | Location map of the Silicate Carbon Frontier prepurchase CDR project.

Project scale

The project will see 6,500 t of carbonate-bearing materials applied to ~650 ha of arable farmland in southeast Ireland in 2025. Material will be provided by Silicate Carbon's longstanding partner, Roadstone, which is part of the CRH group, one of the world's largest building materials companies. Farmland will be provided by Silicate Carbon's existing agricultural partners, including large-scale agricultural companies and independent farmers. Silicate Carbon will undertake all associated sampling and measurement activity in-house and with external laboratory facilities.

The project will run for a minimum of two years, with the Frontier contract being completed before the end of 2027. From long-term (>4-year) field-scale research undertaken by Silicate Carbon, the in-field dissolution kinetics of carbonate feedstock materials are well understood. This knowledge paired with the team's field selection tool and MRV capabilities will ensure maximum H_2CO_3 dissolution of the feedstock material, creating large and consistent increases in HCO_3^- , while measuring rainfall at multiple sites to account for spatial gradients in water fluxes. Importantly, considerations of additionality above a BAU baseline will be applied during the field selection and CDR calculation phases, ensuring that CDR is both robust and additional.

Frontier cost criteria

The primary costs of Silicate Carbon's project are EW feedstock materials, field sampling, and laboratory analyses. Based on Silicate Carbon's direct sampling of HCO_3^- in field soil waters, as opposed to stoichiometric assessments of maximum CDR potential t^{-1} , efficient CDR can be achieved at scale that can comfortably meet the $<\$100 \text{ t}^{-1}$ CDR threshold. Both material and material application costs are unlikely to fall as the solution scales due to the embedded costs involved in quarrying, processing, transporting and applying materials to farmland. Instead, cost efficiencies are achieved through high CDR efficiencies per t^{-1} material applied, and low cost, robust MRV.

Field sampling and laboratory analysis costs will decline over time as detailed field sampling becomes increasingly redundant. Silicate Carbon continues to refine its models based on robust soil water and

other field data, which will enable low-cost CDR measurement at scale, further reducing CDR costs to $< \$100 \text{ t}^{-1}$ CDR.

Frontier scale criteria

To expand EW to achieve Gt y^{-1} scale, EW must deliver clear, tangible benefits to farmers. Limestone has been applied to fields to counteract the soil acidifying effects of nitrogenous fertilizers for centuries, so the long-term effects of limestone are well understood^{21,22}. By optimizing soil pH amendment for CDR, additional value can be created for farmers. By advancing EW as a CDR strategy in an agricultural context, farm resilience from increased agricultural productivity and profitability can be enhanced.

Moreover, a limited fraction of the global 6.6 Gt of limestone quarried each year is used for soil pH amendment. Though estimates for milled limestone are not broadly available, $\sim 300 \text{ Mt}$ are assumed to be applied to farmland to correct soil pH annually. Given the extent of agricultural land that is somewhat acidic (1.7 billion ha), Silicate Carbon proposes to greatly expand the application of limestone to farmland. This increase in material application will not only create additional CDR but will unlock further farm productivity enhancements by optimizing soil pH for crop productivity.

Measuring CDR

Measurement is foundational to the success of EW as a CDR pathway. At a minimum, it is necessary to measure the HCO_3^- alkalinity of soil porewaters and agricultural runoff. It is also necessary to measure other parameters of water (i.e., T, pH, cations, and anions) to assess the thermodynamic saturation state with respect to calcite and other carbonate minerals. As well as doing these water measurements, Silicate Carbon will analyze soils and biological materials to comprehensively constrain C mass-balance, as well as mass-balances for Ca and Mg, which affect HCO_3^- via charge balance. Among other goals and objectives, this project will advance high-precision isotopic methods as novel measurement tools to trace carbon and cation sources and pathways, and will constrain the role of cation exchange phenomena, which play a critical role in the release and transport of HCO_3^- .

Suction-cup lysimeters will be installed to collect shallow and deep soil waters across the project site (Figure 4). In-situ measurements will include water pH and T, as well as crop yield and soil gas (CO_2 , N_2O) effluxes (Figure 4). Intact soil cores will be collected before, during, and after amendments (Figure 4). Laboratory measurements will include elemental and isotopic analyses of waters, crops, soils, and amendment materials. In addition, soils will be assayed to quantify microbial community compositions and functions, as well as organic carbon concentrations, pH, and cation exchange capacity (CEC). Empirical data will be used to calibrate a first-principles 1D reactive-transport model for quantifying water-rock interactions, predicting CDR efficacy over time, and optimizing future field trials. The modeling effort will address uncertainty with statistical methods.

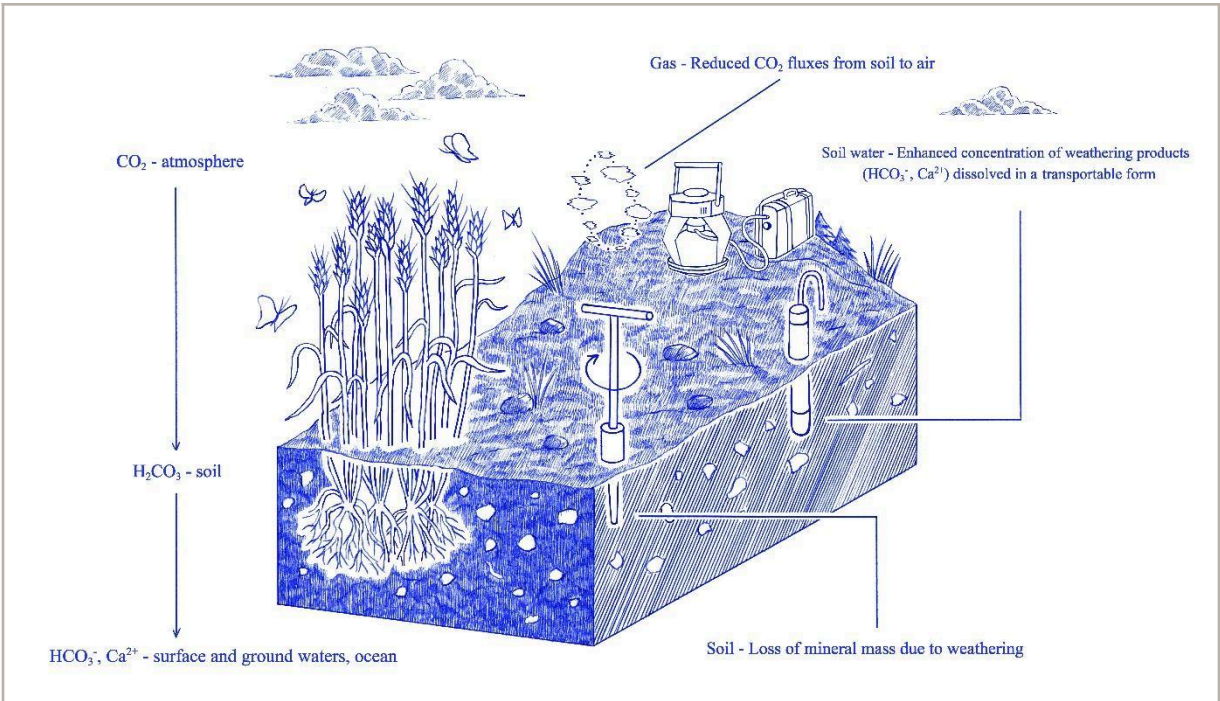


Figure 4 | Measurements employed by Silicate Carbon during the Frontier prepurchase CDR project.

Conclusion

The anticipated impact of the proposed project is sizable. Diversifying EW with new materials and to new geographies, while furthering the development of high-quality and transparent CDR measurement is essential for imbuing farmers, investors, government officials, and the public with confidence in EW as an impactful CDR pathway. Silicate Carbon’s efforts during this Frontier project will further help to develop, refine, and hone high-quality and transparent measurement, while demonstrating the rapid CDR and scaling potential of a novel EW feedstock material, limestone.

- c. **Risks:** What are the biggest risks and how will you mitigate those? Include technical, project execution, measurement, reporting and verification (MRV), ecosystem, financial, and any other risks. 500-1000 words

Overview

Though several academic papers^{15,23,24} demonstrate the immense CDR potential of EW, the science underpinning EW, as well as its economics, is still at a nascent stage. Demonstrating CDR through EW at a minimum requires evidence of increased HCO_3^- production and export to the ocean, which, to date, has not been evident in the EW literature^{19,25}.

Technical

- **Strong acid weathering of feedstock material**
 - **Description:** A potential but manageable risk is that in some soils, weathering might be driven by strong acids, such as nitric acid (HNO_3) not H_2CO_3 , the neutralisation of which sequesters soil and therefore atmospheric CO_2
 - **Solution:** Silicate Carbon research (unpublished) demonstrates that strong acid weathering can be minimal and mitigated
 - **Solution:** Soil water chemistry provides information on the anionic species charge balancing released cations in amended fields. Soil water samples will be analyzed using Silicate Carbon's Metrohm IC, and form part of the MRV package
- **In situ CO_2 loss pathways**
 - **Description:** The net impact of EW on soil organic carbon (SOC) stocks is largely unknown, and experimental results to date are variable²⁶
 - **Solution:** Long-term continuous measurement of CO_2 fluxes, such as the trials Silicate Carbon is hosting in its U.S. Midwest base offer the most robust means for quantifying these losses and building appropriate post-amendment SOC models
- **Downstream CO_2 loss pathways**
 - **Description:** HCO_3^- produced by EW could be lost as CO_2 during riverine transport away from agricultural sites and to oceans. Possible loss pathways include, but are not limited to, calcite precipitation and CO_2 evasion, both of which have been extensively documented to occur in diverse fluvial systems²⁷⁻³⁰
 - **Solution:** Efforts to constrain HCO_3^- losses have to-date been model based^{14,31}. Silicate Carbon's Frontier prepurchase project will sample stream water from low-order streams draining the deployment region (Figure 3) accounting for spatial and seasonal gradients

Project execution

- **Feedstock material supply**
 - **Description:** Adequate supply of material suitable for application to farmland is essential for CDR with EW. Consideration of mineral composition, weathering kinetics, distance to farmland, among other factors, is important
 - **Solution:** Silicate Carbon has robust partnerships with companies owning and operating limestone quarries around the world. Importantly, the quarries supplying material to the proposed Frontier prepurchase project have electric milling infrastructure installed, enabling low-cost and low CO_2 emissions quaternary processing, and are proximal to field sites, further lowering process emissions
- **Suitable farmland acquisition**
 - **Description:** Access to suitable farmland (target pH and rainfall, for example) is paramount to EW's success as a CDR pathway
 - **Solution:** Ireland represents a tremendous opportunity for carbonate EW as its base lithology, soils, and predominant weather systems maintain optimal conditions for carbonate EW. Moreover, carbonate minerals exhibit retrograde solubility, meaning that they weather efficiently in temperate climates, further supporting their adoption as an EW material in mid-latitude agricultural regions

Measurement reporting and verification

- **Constraining delta HCO_3^-**
 - **Description:** The temporal signature of aqueous phases can vary with rainfall amount and intensity, affecting dissolution kinetics. Temporal and scale (field, catchment etc.)

- considerations should be included in the MRV protocol to mitigate these challenges
 - o **Solution:** Silicate Carbon has run large (450-ha) projects with a weekly soil water sampling cadence. The team has developed insights on project design and sampling frequency that mitigate commonly held concerns with soil water sampling
 - o **Solution:** Silicate Carbon has developed novel methods to sample soil pore water efficiently and at scale. These proprietary methods will be employed in the Frontier prepurchase project and will help to reduce sampling costs as the solution scales
 - o **Solution:** Silicate Carbon will sample soil waters in complementary studies to the proposed project to monitor soil water chemistry in a hydrologically isolated field, and at the catchment scale. These data will further demonstrate the efficacy and utility in direct HCO_3^- sampling in water chemistry as an effective MRV method for EW
- **Constraining GHG efflux**
 - o **Description:** Strong acid weathering of carbonate and silicate rocks can release CO_2 —many basalts contain trace amounts of calcite between ~0.4-3.0%^{32,33}. Increased soil-pH through liming can reduce N_2O emissions³⁴. As N_2O is a more potent GHG than CO_2 , reduced emissions could be an important co-benefit of EW
 - o **Solution:** CO_2 and N_2O fluxes will be measured weekly using LI-COR LI-7810 and LI-7820 trace gas analyzers, respectively, and 8200-01S smart chambers installed in research plots in the project area. Additionally, Silicate is using new technology, such as CO_2 isotope flux measurements, to trace the origin of CO_2 emitted from the soil and to determine if the CO_2 derives from limestone weathering, organic matter respiration, or some combination thereof

Ecosystem

- **Resource consumption**
 - o **Description:** EW consumes rocks and material extraction and pulverization is required to enhance weathering above background levels^{13,35,36}
 - o **Solution:** Compared to other CDR approaches, EW demonstrates significant advantages: it requires half the energy of direct air capture plants and uses 10-100 times less water than direct air capture, bioenergy, and biochar pathways³⁷
 - o **Solution:** The specific crushing energy of limestone is 1.65-1.93 kJ kg⁻¹, lower than basalt's 2.56-4.09 kJ kg⁻¹³⁸. Annual limestone production is estimated to be more than 6.6 Gt¹², accounting for 9% of the world production of mineral raw materials (~44 Gt y⁻¹), meaning that expansion for EW is attainable with minimal expansion to current limestone production
- **Soils, plants and downstream waters**
 - o **Description:** The leachability of toxic trace elements is a major concern for ultramafic and mafic materials²³
 - o **Solution:** Unlike mafic and ultramafic materials, limestone primarily comprises CaCO_3 and lacks significant concentrations of toxic metals. Limestone amendments improve soil pH, reducing the solubility and mobility of heavy metals, which are more bioavailable and toxic in acidic conditions

Financial

- **Security of sufficient debt, equity and/or project financing**
 - o **Description:** EW has low capex costs in comparison to engineered pathways; however, cashflow burdens placed on EW companies could be unsustainable if revenue generation (through CDR credit sales) is extended beyond financeable

- timelines
 - o **Solution:** Limestone dissolution by H_2CO_3 is rapid (<4 years), meaning that revenue can meet costs with existing financial instruments and structures
- d. **Proposed offer to Frontier:** Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons)
(should be net volume after taking into account the uncertainty discount proposed in 5c)

780 t¹ CDR

Delivery window
(at what point should Frontier consider your contract complete? Should match 2f)

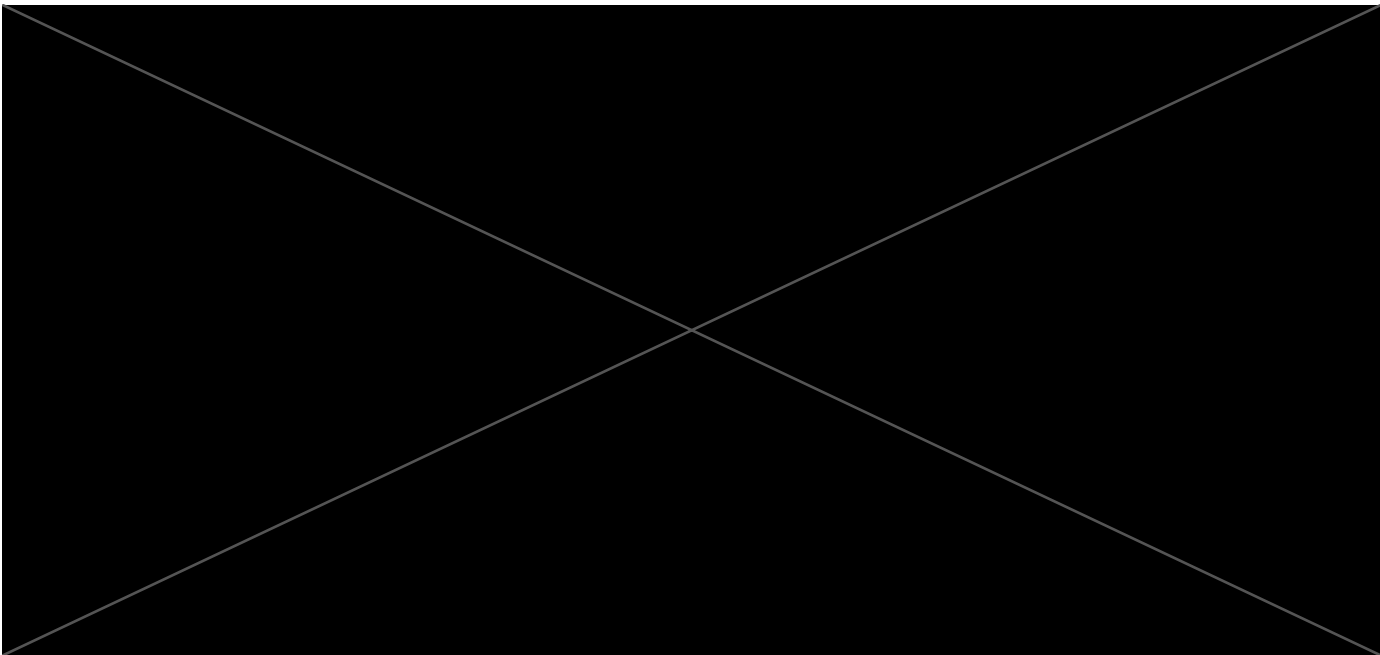
Q4 2027

Levelized cost (\$/ton CO₂)
(This is the cost per ton for the project tonnage described above, and should match 6d)

\$550 t¹ CO₂

Levelized price (\$/ton CO₂)³
(This is the price per ton of your offer to us for the tonnage described above)

\$600 t¹ CO₂



³ This does not need to exactly match the cost calculated for “This Project” in the TEA spreadsheet (e.g., it’s expected to include a margin and reflect reductions from co-product revenue if applicable).